

JOSEP LOPEZ CAMUÑAS

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EDUCATION

Universitat Oberta de Catalunya (UOC) 2022 – present (expected 2027)
Ph.D. in Computer Science (Advised by Prof. Agata Lapedriza)

- Research focus: Multimodal Learning, Emotion Understanding, Explainable AI, and Social Robotics.

Universitat Autònoma de Barcelona (UAB) 2021 – 2022
M.Sc. in Computer Vision

- Thesis: *Non-verbal Synchrony Analysis in Dyadic Interaction*. Grade: A.

Universitat de Barcelona (UB) 2016 – 2020
B.Sc. in Computer Science

INDUSTRY AND RESEARCH EXPERIENCE

Never Home Alone (NHoA) 2022 – 2025
Researcher — Affective and Social Robotics Barcelona, Catalunya, Spain

- Developed intelligent robotics for elderly care by designing emotion-perception systems and collecting user datasets.

MIT CSAIL — Torralba Lab Feb 2025 – June 2025
PhD Visiting Student Cambridge, Massachusetts, USA

- Researched Automated Interpretability Agents for VLMs with Prof. A. Torralba; adapted MAIA for open-source models.

Institute for Experiential AI (EAI) — Northeastern University Mar 2024 – June 2024
Feb 2025 – June 2025
PhD Visiting Student Boston, Massachusetts, USA

- Conducted research on improving visual emotion and affect representations with Prof. A. Lapedriza and Dr. Y. Zhu.

Universitat Oberta de Catalunya (UOC) Oct 2021 – June 2022
Research Intern in Computer Vision Barcelona, Catalunya, Spain

- Developed facial and body analysis methods and performed computational modeling for dyadic interactions.

DMD.Solutions
Data Scientist

Feb 2020 – Nov 2021
Barcelona, Catalunya, Spain

- Developed and evaluated predictive maintenance models for automated aircraft fault prediction.

HONORS

- **PhD Fellowship**, Universitat Oberta de Catalunya (UOC), 2022 – present.

ACADEMIC SERVICE

- **Reviewer:** CVPR 2025, NeurIPS 2025, WACV 2025, CIARP 2025 | **Organization:** KDD 2024 (Volunteer).
- **Supervision:** Diana Tat (Master Thesis) | **Teaching Assistant:** UPC - Advanced Computer Vision.

RESEARCH PUBLICATIONS

- 2026 **Semantic Justifications for Multiple Interpretations of Visual Emotion Perception with Open Vocabulary (*Under Review*)**. Josep Lopez Camuñas, Cristina Bustos, Yanjun Zhu, Cristian Gutierrez, Agata Lapedriza.
- 2026 **CodeFace: Explainable 3D Face Dynamics from Multi-modal Glasses via Vector-Quantized Priors (*Under Review*)**. Zuyu Guo, Yanjun Zhu, Josep Lopez Camuñas, et al.
- 2025 **Automated Detection of Visual Attribute Reliance with a Self-Reflective Agent**. Christy Li, Josep Lopez Camuñas, Jake Thomas Touchet, Jacob Andreas, Agata Lapedriza, Antonio Torralba, Tamar Rott Shaham. NeurIPS 2025.
- 2025 **OpenMAIA: A Multimodal Automated Interpretability Agent Based on Open-Source Models**. Josep Lopez Camuñas, Christy Li, Tamar Rott Shaham, Antonio Torralba, Agata Lapedriza. Mechanistic Interpretability Workshop at NeurIPS 2025.
- 2025 **Experimenting with Affective Computing Models in Video Interviews with Spanish-Speaking Older Adults**. Josep López Camuñas, Cristina Bustos, Yanjun Zhu, Raquel Ros, Agata Lapedriza. WACV 2025 Workshops (CV4Small).

SKILLS

Programming: Python, C++, C.

AI & Deep Learning: PyTorch, HuggingFace, NumPy, Pandas.

MLOps & Infrastructure: Docker, AWS, FastAPI, vLLM, Linux, Git.

HPC & Scaling: Distributed Training (DDP, Accelerate), Multi-GPU/Node, NVIDIA NGC.

Domain Expertise: Multimodal Automated Interpretability Agents, Vision-Language Models (VLMs), Affective Computing.

Languages: Spanish (native), Catalan (native), English (fluent).